

Estimating Galaxy Redshift in Radio-Selected Datasets using Machine Learning

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Abstract

All-sky radio surveys are set to revolutionise the field with new discoveries. However, the vast majority of the tens of millions of radio galaxies won't have the spectroscopic redshift measurements required for a large number of science cases. Here, we evaluate techniques for estimating redshifts of galaxies from a radio-selected survey. Using a radio-selected sample with broadband photometry at infrared and optical wavelengths, we test the k -Nearest Neighbours (kNN) and Random Forest machine learning algorithms, testing them both in their regression and classification modes. Further, we test different distance metrics used by the kNN algorithm, including the standard Euclidean distance, the Mahalanobis distance and a learned distance metric for both the regression mode (the Metric Learning for Kernel Regression metric) and the classification mode (the Large Margin Nearest Neighbour metric). We find that all regression-based modes fail on galaxies at a redshift $z > 1$. However, below this range, the kNN algorithm using the Mahalanobis distance metric performs best, with an $\eta_{0.15}$ outlier rate of 5.85%. In the classification mode, the kNN algorithm using the Mahalanobis distance metric also performs best, with an $\eta_{0.15}$ outlier rate of 5.85%, correctly placing 74% of galaxies in the top $z > 1.02$ bin. Finally, we also tested the effect of training in one field and applying the trained algorithm to similar data from another field and found that variation across fields does not result in statistically significant differences in predicted redshifts. Importantly, we find that while we may not be able to predict a continuous value for high-redshift radio sources, we can identify the majority of them using the classification modes of existing techniques.

Keywords: methods: analytical; techniques: photometric; galaxies: photometry; galaxies: high-redshift

1. Introduction

New radio telescopes are set to revolutionise the radio astronomy regime, with the Evolutionary Map of the Universe (EMU) project to be completed on the Australian Square Kilometre Array Pathfinder (ASKAP; Johnston et al., 2007, 2008) telescope in particular set to increase the number of known radio sources from ~ 2.5 million (Norris, 2017) to ~ 70 million (Norris et al., 2011).

For most aspects of science, knowledge of an astronomical source's redshift is an essential indicator of the distance and age of the source. Ideally, this redshift is measured directly using spectroscopy. However, even with modern advances — including Multi-Object Spectroscopy (MOS) which can allow hundreds to thousands of redshifts to be measured at once — deep spectroscopic surveys still fail to yield reliable redshifts from 30-60% of measured spectra (Newman et al., 2015). Currently, the Sloan Digital Sky Survey (SDSS) has measured ~ 4.8 million spectroscopic redshifts — as of the 16th data release (Ahumada et al., 2020)¹ — and the 2dF Galaxy Redshift Survey (2dFGRS;

Lewis et al., 2002) measured $\sim 250\,000$ spectroscopic redshifts over its 5 year project². In the future, the Wide Area Vista Extragalactic Survey (WAVES) survey — expected to provide a further 2.5 million spectroscopic redshift measurements (Driver et al., 2016) — will increase the number of spectroscopically measured sources. However, this will still be significantly short of the expected ~ 70 million sources detected by the EMU project (even if all of these newly measured galaxies were exclusively selected from the EMU survey).

Photometric template fitting (Baum, 1957) is able to estimate the redshift (hereafter z_{photo}) of a source, as well as ancillary data like galaxy classification. Photometric template fitting is completed by comparing the Spectral Energy Distribution (SED), measured across as many different wavelengths as possible, to templates constructed with astrophysical knowledge, or prior examples. Figure 1 shows the 31 filter bands available in the COSMOS evolution Survey (COSMOS) field and an example SED used by Ilbert et al. (2009) to achieve redshift accuracies of $\sigma_{\Delta z/(1+z_s)} = 0.012$.

Machine Learning (ML) has also been applied to the problem, with the k -Nearest Neighbours (kNN) algorithm

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¹<https://www.sdss.org/dr16/scope/>

²<http://www.2dfgrs.net/Public/Survey/statusfinal.html>

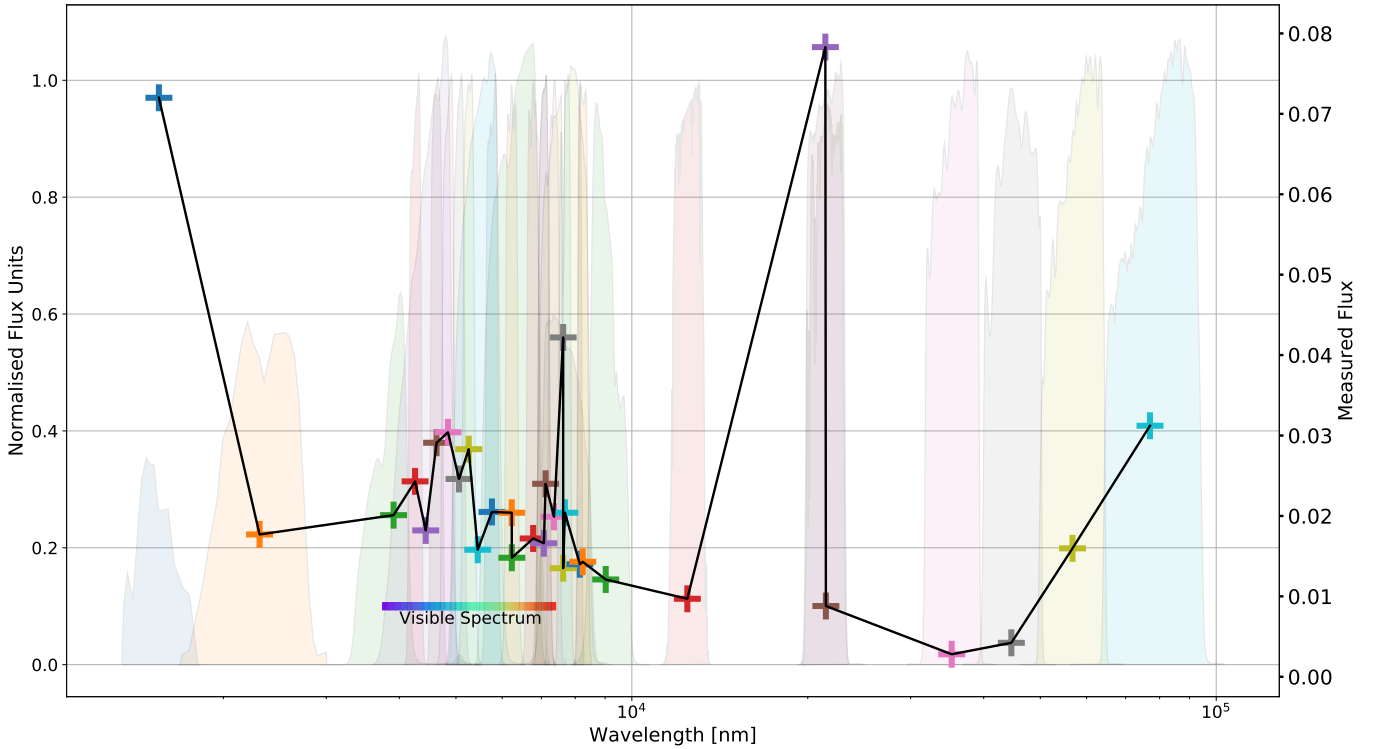


Figure (1) The SED of an extragalactic source from the COSMOS survey (Object ID: 358521). The background shows the filter coverage of the 31 filters available in the COSMOS field (defined in Ilbert et al. (2009)). The “+” symbols in the foreground represent the galaxy’s photometric measurements in each band, and are coloured to match the corresponding filter in the background.

(Ball et al., 2007, 2008; Oyaizu et al., 2008; Zhang et al., 2013; Kügler et al., 2015; Cavioti et al., 2017; Luken et al., 2019, 2021), Random Forest (RF; Cavioti et al., 2012, 2015; Hoyle, 2016; Sadeh et al., 2016; Cavioti et al., 2017; Pasquet-Itam and Pasquet, 2018), and neural networks (Firth et al., 2003; Tagliaferri et al., 2003; Collister and Lahav, 2004; Brodwin et al., 2006; Oyaizu et al., 2008; Hoyle, 2016; Sadeh et al., 2016; Curran, 2020; Curran et al., 2021) being among the more widely used algorithms. Some recent studies utilise Gaussian Process (GP; Duncan et al., 2018a,b; Duncan et al., 2021), and deep learning using the original images at different wavelengths, as opposed to the photometry extracted from the image (D’Isanto and Polsterer, 2018).

However, few of these solutions are appropriate for the large-scale radio surveys being conducted. While photometric template fitting provides a theoretically ideal solution, the quality of data required to make it highly accurate will not be available for the majority of sources in the all-sky EMU survey, since only all-sky photometric data will be available for most sources. An example of the broad-band photometry used by this work is shown in Figure 2, which shows both a likely Active Galactic Nucleus (AGN; top), and Star Forming Galaxy (SFG; bottom).

Additionally, Norris et al. (2019) has shown that photometric template fitting performs poorly on radio-selected data sets (which are typically dominated by AGN), possibly because the majority of templates are unable to dif-

ferentiate between emission from the AGN and emission from the galaxy itself (Salvato et al., 2018).

The ML based methods have mainly focused on optically-selected datasets, with most work drawing on the SDSS photometric and spectroscopic samples, often restricting the redshift range to $z < 1$, or using datasets containing only one type of object — e.g. the SDSS Galaxy or Quasi-Stellar Object (QSO) catalogs. While this kind of testing is entirely appropriate for non-radio selected samples, Norris et al. (2019) have shown that assumptions in the optical regime may not be valid in the radio regime, for two reasons. First, the redshift distribution of radio-selected sources is quite different from that of optically-selected sources, as shown in Figure 3. Second, radio-selected sources are often dominated by a radio-loud AGN which is poorly represented in optically-selected templates and training sets.

Finally, we should note that in support of the science goals of the EMU Project, we aim to minimise the number of estimations that catastrophically fail, rather than minimise for accuracy as most other works do.

The overall contributions of this study include:

- An in-depth study of the simple k -Nearest Neighbours algorithm, including the investigation of typically ignored distance metrics for the estimation of radio galaxy redshift;
- A comparison with the widely used Random Forest

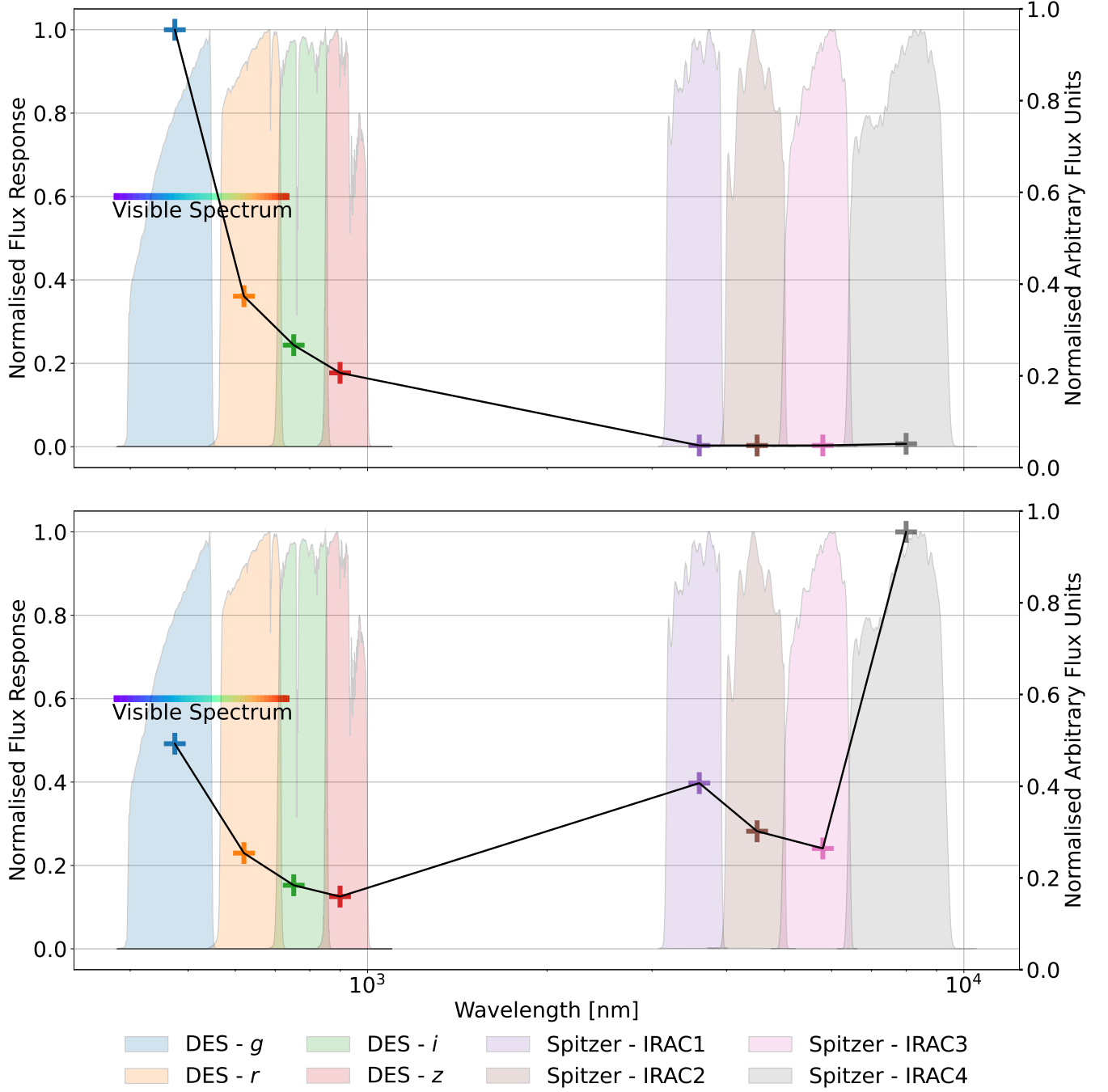


Figure (2) The SED of the extragalactic source ATLAS3_J033551.0283345C (top; likely AGN) and ATLAS3_J033519.0273708C (bottom; likely SFG), in similar style to Figure 1. The background shows the filter coverage used by this work, with the “+” in the foreground representing the measured photometry at each band, taken from the data used in this study.

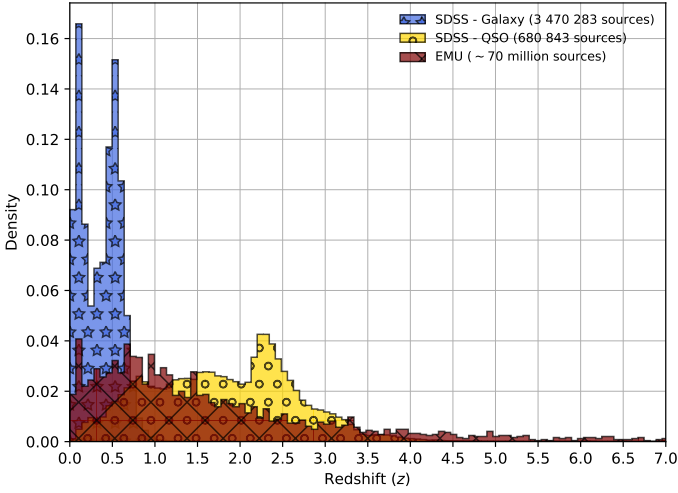


Figure (3) The normalised distributions of redshift in the SDSS Galaxy and QSO surveys, compared with the expected redshift distribution of the EMU survey, modelled by the Square Kilometre Array Design Survey (SKADS; Levrier et al., 2009).

algorithm;

- An analysis of whether similarly observed fields can be used as training samples for alternate fields;
- An exploration into the effectiveness of regression vs classification modes in estimating high redshift galaxies, given the highly unbalanced nature of the training sample (noting that while classification modes might be able to identify high-redshift galaxies, it will not be able to estimate its actual redshift).

1.1. Formal Problem Statement

In this article, we investigate the problem of modelling the redshift of a source with respect to known measurements of the source. This can be a regression problem, where we attempt to model the function $r_i = f(\vec{x}_i)$, such that the redshift of source i , $r_i \in \mathbb{R}^+$ and $\vec{x}_i$ the known measurements of source i are obtained from a catalogue with various domains. We can also model the redshift of a source as $c_i = g(\vec{x}_i)$, where c_i is a class representing a specific domain of redshift. In the following section we describe the catalogues containing the known source information $\vec{x}_i$ and the variables that are used for this representation.

2. Data

This work uses data from the Australia Telescope Large Area Survey (ATLAS; Norris et al., 2006; Franzen et al., 2015) radio continuum catalogue, providing a 1.4 GHz flux density measurement of 4780 unique sources. The ATLAS survey covers two regions of the sky – the extended Chandra Deep Field South (eCDFs) and the European Large Area ISO Survey–South 1 (ELAIS-S1), both to a depth of $\sim 15\mu\text{Jy}$, and was completed as a first-look at what the

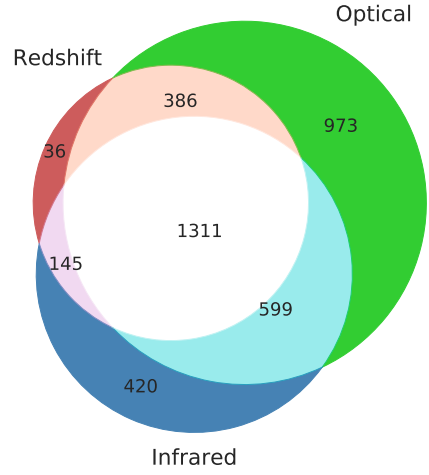


Figure (4) A Venn diagram of the number of radio sources in the ATLAS dataset, with a spectroscopic redshift (provided by the OzDES), optical magnitudes at g , r , i , and z bands (provided by the DES) and infrared fluxes at 3.6, 4.5, 5.8 and 8.0 μm (provided by the SWIRE survey)

EMU survey may provide. This survey was cross-matched by Swan (2018) with the Australian Dark Energy Survey (OzDES; Yuan et al., 2015; Childress et al., 2017; Lidman et al., 2020), Dark Energy Survey (DES; Dark Energy Survey Collaboration et al., 2016) and Spitzer Wide-Area Infrared Extragalactic Survey (SWIRE; Lonsdale et al., 2003) surveys, providing spectroscopic redshift measurements, g , r , i and z optical magnitudes, and 3.6, 4.5, 5.4 and 8.0 μm infrared flux measurements respectively.

This work used only those ATLAS sources with measured photometry in all provided optical and infrared bands, creating a final dataset containing 1311 sources with complete photometric coverage. Specifically, in both fields used in this work, we used the g , r , i and z optical magnitudes from DES, and 3.6, 4.5, 5.4 and 8.0 μm infrared flux measurements from the SWIRE survey, as shown in Figure 4. The redshift distribution of the final collated dataset is presented in Figure 5.

To prevent the different methods tested from being dominated by single features with wide variation, all features were standardised using Equation 1, setting the feature mean to 0, and the feature variance to unit variance.

$$z_i = \frac{x_i - \bar{x}}{s_x}, \quad (1)$$

where $\bar{x}$ is the mean of the sample of variable x and s_x is its standard deviation.

During testing, we also examined the use of “colours” rather than optical magnitudes, and taking the log of the radio and infrared fluxes to better distribute the data. Colours are typically used instead of magnitudes to remove the brightness- and redshift-dependent nature of magnitudes, replacing them by the difference between the magnitudes that is dependent only on the SED.

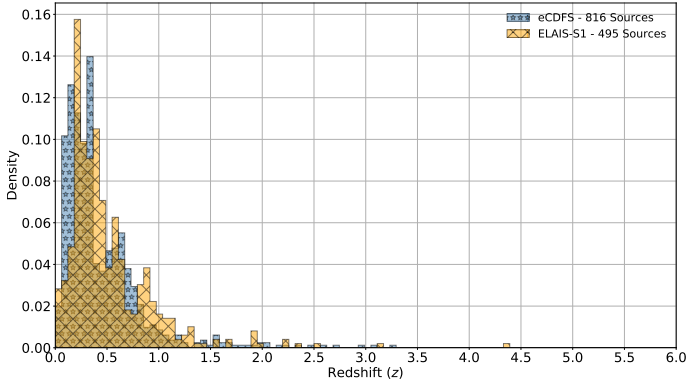


Figure (5) The redshift distribution of sources in the ATLAS dataset, broken down by field (ELAIS-S1 and eCDFs). The overall mean redshift is $z = 0.47$, ELAIS-S1 mean is $z = 0.51$ and eCDFs mean is $z = 0.44$

2.1. Data Partitioning

All machine learning methods require the main dataset to be partitioned into multiple subsets, with the subsets set aside for training the model, validating hyperparameters, and testing the model. In this work, we split our data into two datasets – training and testing, with the hyperparameters being validated using k -fold cross validation on the training set. We partition our data differently for the three tests we complete:

- Typical random split, with 70% of the data split off as the training set, leaving the remaining 30% as the test set.
- The entire ELAIS-S1 field as the training set, leaving the eCDFs field as the test set.
- The entire eCDFs field as the training set, leaving the ELAIS-S1 field as the test set.

In addition to the above partitioning, in order to facilitate the use of the classification modes of our algorithms tested, we quantise the redshift values into 15 redshift bins (defined in Table 1), with equal numbers of sources in each in order to be able to predict a uniform distribution with using a matching distribution. **The median of the spectroscopic redshifts within the bin is chosen as the redshift to predict.**

3. Methods

3.1. kNN

The kNN algorithm (Cover and Hart, 1967) computes a similarity matrix between all sources based on the catalogue’s photometric measurements, and a given distance metric (the set of distance metrics tested in this work are explained in Subsections 3.1.1, 3.1.2, and 3.1.3). Once the similarity matrix is constructed, the kNN algorithm finds the k (Hereafter k_N) most similar sources with measured redshifts (where k_N is optimised using cross validation),

Table (1) Details of the bins used for the classifications tests, outlining the lower edge, upper edge and predicted value of each bin (where the predicted value is the median of the spectroscopic redshifts from within the bin).

Bin Number	Lower Bound	Upper Bound	Median Value
1	0	0.10	< 0.1
2	0.10	0.15	0.12
3	0.15	0.19	0.17
4	0.19	0.22	0.21
5	0.22	0.26	0.24
6	0.26	0.29	0.27
7	0.29	0.32	0.31
8	0.32	0.35	0.34
9	0.35	0.41	0.38
10	0.41	0.50	0.46
11	0.50	0.58	0.54
12	0.58	0.66	0.62
13	0.66	0.80	0.73
14	0.80	1.02	0.91
15	1.02	4.33	> 1.02

and takes either the mean value (for regression) or the mode class (for classification) of the sources as the estimated redshift for each source. A simple illustration of the kNN algorithm is shown in Figure 6

The value of k_N used in the kNN in this work is optimised using k -fold (where k is hereafter k_f and is set to 10 for this work) cross-validation. k_f -fold cross-validation randomly splits the data into k_f roughly even groups, iterating through using $k_f - 1$ groups as the training set and validating on the remaining group, until every group has been used in testing, with the average error used as the error for that value of k_f . This work tested all integer values of k_N for the kNN algorithm between 2 and 20 ($k_N \in \{2, \dots, 20\}$).

The kNN algorithm requires a metric to determine which of the observations are neighbours to a given observation. In the next sections we present the set of metrics investigated in the experiment.

3.1.1. Euclidean Distance

Euclidean Distance is the simplest, and most widely used distance metric in literature, defined in Equation 2:

$$d(\vec{p}, \vec{q}) = \sqrt{(\vec{p} - \vec{q})^T (\vec{p} - \vec{q})}, \quad (2)$$

where $d(\vec{p}, \vec{q})$ is the Euclidean distance between two feature vectors $\vec{p}$ and $\vec{q}$. In this case the vectors contain the measured photometry of two galaxies.

3.1.2. Mahalanobis Distance

The Mahalanobis distance metric (Mahalanobis, 1936) normalises the variance and covariance of the input features by transforming the features using the inverse of the covariance matrix. For uncorrelated input features, the

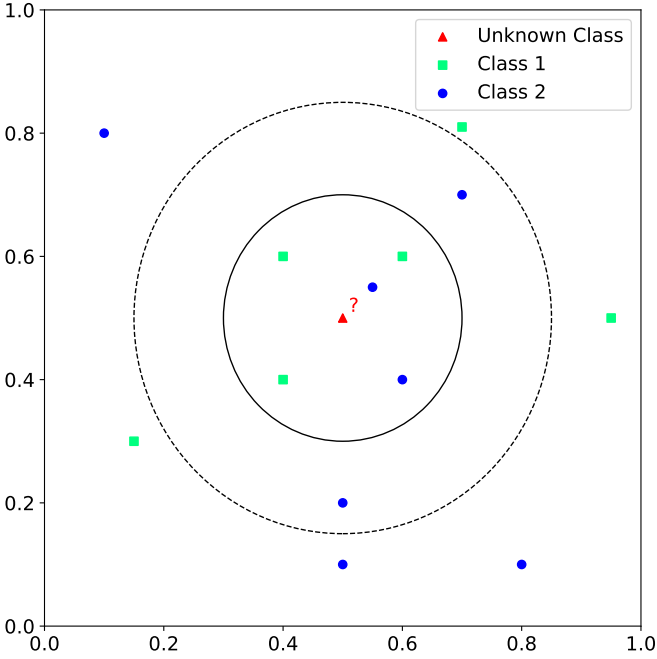


Figure (6) A pictorial representation of the k -Nearest Neighbours algorithm. If k_N is chosen to be 5 (denoted by the solid circle), by taking the mode of the k_N nearest neighbours, the red triangle would be assigned a class of Green Square. If the value of k_N was changed to 7 (denoted by the dashed circle), the mode of the k_N nearest neighbours would be the Blue Circle. Modified from: Ajanki, Antti (2007)

Mahalanobis distance is equal to the scaled Euclidean distance, but for correlated features, it generalises the idea of Euclidean distance to take account of the covariance. The Mahalanobis distance is defined in Equation 3:

$$d(\vec{p}, \vec{q}) = \sqrt{(\vec{p} - \vec{q})^T S^{-1} (\vec{p} - \vec{q})}, \quad (3)$$

where $d(\vec{p}, \vec{q})$ is the Mahalanobis distance between two feature vectors $\vec{p}$ and $\vec{q}$, and S is the covariance matrix.

3.1.3. Learned Distance Metrics

Conceptually, for best results the distance metric used should take into consideration the shape and structure of the data. Towards this end, we can generalise Equations 2, and 3 to Equation 4, noting that the $\mathbf{M}$ matrix can be any positive semi-definite matrix. The Identity matrix is used for Euclidean Distance, and the S^{-1} matrix is used in the Mahalanobis. We can, however, go one step further, and attempt to learn an $\mathbf{M}$ matrix that can better warp the feature space so that observations with similar redshift are measured as close, while observations with different redshift are measured as distant.

$$d(\vec{p}, \vec{q}) = \sqrt{(\vec{p} - \vec{q})^T \mathbf{M} (\vec{p} - \vec{q})}, \quad (4)$$

where $d(\vec{p}, \vec{q})$ is the distance between two feature vectors $\vec{p}$ and $\vec{q}$.

For our regression tests, we used the Metric Learning for Kernel Regression (MLKR) distance metric, which performs a supervised Principle Component Analysis (Weinberger and Tesauro, 2007). The MLKR distance metric begins by decomposing the $\mathbf{M}$ matrix from Equation 4 using Equation 5:

$$\mathbf{M} = \mathbf{A}^T \mathbf{A} \quad (5)$$

Using Equation 5, Equation 4 can be expressed as the modified Euclidean Distance metric in Equation 6:

$$d(\vec{x}_i, \vec{x}_j) = \|\mathbf{A}(\vec{x}_i - \vec{x}_j)\|^2 \quad (6)$$

Matrix $\mathbf{A}$ is optimised using Gradient Descent, using Equation 7:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{A}} = 4\mathbf{A} \sum_i (\hat{y}_i - y_i) \sum_j (\hat{y}_j - y_j) k_{ij} \vec{x}_{ij} \vec{x}_{ij}^T, \quad (7)$$

where $\hat{y}_i$ is defined in Equation 8, k_{ij} is a Gaussian kernel defined in Equation 9, and $\vec{x}_{ij} = (y_i - \hat{y}_i)^2$.

$$\hat{y}_i = \frac{\sum_{j \neq i} y_j k_{ij}}{\sum_{j \neq i} k_{ij}} \quad (8)$$

$$k_{ij} = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{d(\vec{x}_i, \vec{x}_j)}{\sigma^2}} \quad (9)$$

The loss function being minimised is a simple squared difference, defined in Equation 10:

$$\mathcal{L} = \sum_i (y_i - \hat{y}_i)^2 \quad (10)$$

For our Classification tests, the Large Margin Nearest Neighbour (LMNN; Weinberger et al., 2006) learned distance metric was used. The LMNN distance metric finds a transformation for the data that maximises the distance between different classes, and minimises the distance between similar classes. The loss function the LMNN algorithm optimises is defined in Equation 11:

$$\begin{aligned} \epsilon(\mathbf{L}) = & \sum_{ij} \eta_{ij} \|\mathbf{L}(\vec{x}_i - \vec{x}_j)\|^2 \\ & + c \sum_{ijl} \eta_{ij} (1 - y_{il}) [1 + \\ & \quad \|\mathbf{L}(\vec{x}_i - \vec{x}_j)\|^2 - \\ & \quad \|\mathbf{L}(\vec{x}_i - \vec{x}_l)\|^2]_+, \end{aligned} \quad (11)$$

where i, j , and l are individual galaxy feature vectors, $\eta_{ij} \in \{0, 1\}$ describes whether $\vec{x}_i$ is a target of $\vec{x}_j$, c is a positive constant typically chosen through cross-validation and $[z]_+ = \max(z, 0)$ – a hinge function.

An example of the transformation the LMNN algorithm attempts is in Figure 7, using three neighbours.

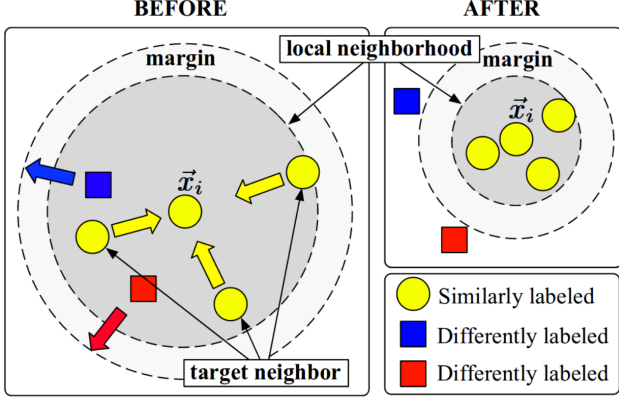


Figure (7) An example of the transformation found by the Large Margin Nearest Neighbour algorithm. The distance between similar classes is minimised, and the distance between different classes is maximised.

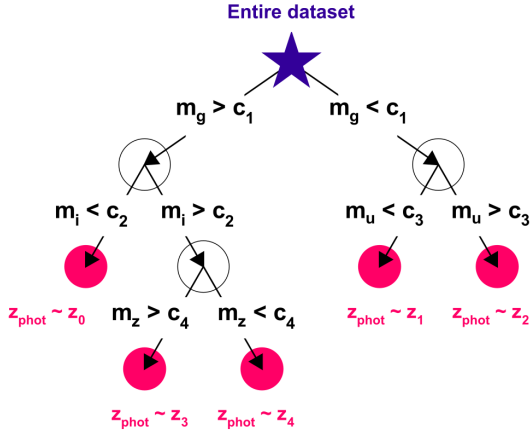


Figure (8) An example of a Decision Tree. Starting at the top with the entire dataset, questions are asked of the data until an outcome is clearly identified.

3.2. Random Forest

For comparison, we have contrasted our results with the popular and well-used RF algorithm. RFs are constructed from bootstrapped Decision Trees (DTs) – an algorithm that partitions the data space so that it can be explored using a tree, shown in Figure 8 (Morgan and Sonquist, 1963; Quinlan, 1987). Each DT finds the impurity at each node (defined in Equation 12) to determine the best features and values to split the data on:

$$G(Q, \theta) = \frac{n_{left}}{N_m} H(Q_{left}(\theta)) + \frac{n_{right}}{N_m} H(Q_{right}(\theta)), \quad (12)$$

where Q is the data at node m , θ is a subset of data, N_m is the number of objects at node m , n_{left} and n_{right} are the numbers of objects on the left and right sides of the split, Q_{left} and Q_{right} are the objects on the left and right sides of the split, and the H function is a impurity function that differs between classification and regression. For Regression, the Mean Square Error is used (defined in

Equation 13), whereas Classification often uses the Gini Impurity (defined in Equation 14).

$$H(X_m) = \frac{1}{N_m} \sum_{i \in N_m} (y_i - \left[\frac{1}{N_m} \sum_{j \in N_m} y_j \right])^2, \quad (13)$$

where N_m is the number of objects at node m and y_i and y_j are the response variables.

$$H(X_m) = \sum_k p_{mk}(1 - p_{mk}), \quad (14)$$

where p_{mk} is the proportion of split m that are class k , defined formally in Equation 15:

$$p_{mk} = \frac{1}{N_m} \sum_{x_i \in R_m} I(y_i = k), \quad (15)$$

where I is the indicator function, identifying the correct classifications.

3.3. Error Metrics

In order to evaluate each of the machine learning models, the prediction error of each model will be assessed. In this section, the set of statistics that measure the error of the models are presented. The primary error metric compared in this work is the $\eta_{0.15}$ outlier rate, defined in Equation 16:

$$\eta_{0.15} = \frac{\text{count}(|\Delta z| > 0.15 \times (1 + z_{spec}))}{\text{Number Of Sources}}, \quad (16)$$

where $\Delta z = z_{spec} - z_{photo}$. The $\eta_{0.15}$ outlier rate is a percentage representing the number of ‘catastrophic failures’ (the percentage of galaxies that have a residual greater than 0.15, scaled with redshift), and is commonly found in literature (Ilbert et al., 2009; Salvato et al., 2009, 2011; Cavaoti et al., 2012; Zitlau et al., 2016; Cavaoti et al., 2017; Jones and Singal, 2017; Mountrichas et al., 2017; Luken et al., 2019; Norris et al., 2019).

We also provide a secondary $\eta_{2\sigma}$ outlier rate in order to provide a statistically sound comparison, defined in Equation 17:

$$\eta_{2\sigma} = \frac{\text{count}(|\Delta z| > 2\sigma)}{\text{Number Of Sources}} \times 100, \quad (17)$$

where $\Delta z = z_{spec} - z_{photo}$, and σ is the standard deviation of the estimated response, defined in Equation 18:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1} (y_i - \hat{y})^2}, \quad (18)$$

where N is the number of observations, y_i is an observation, and $\hat{y}$ is the estimated observation.

As this work is presenting both regression and classification modes of the given algorithms, there are some error metrics that suit regression, and some that suit classification. These metrics are defined in Subsection 3.3.1 and 3.3.2 respectively.

3.3.1. Regression Error Metrics

For the regression tests, three additional error metrics were compared. The first is the Normalised Median Absolute Deviation (NMAD). The NMAD is a similar measure to the standard deviation, and is generally used for non-Gaussian distributions. It is more robust than the standard deviation, as it takes the median of the residuals, improving the resilience to outliers – an issue that can be prominent in redshift estimation due to the highly unbalanced data sets used, and is defined in Equation 19:

$$\sigma_{\text{NMAD}} = 1.4826 \times (\text{median}(|X_i - \text{median}(X)|)), \quad (19)$$

where NMAD is the Normalised Median Absolute Deviation, X is a vector of residuals from which X_i is taken.

The R^2 Coefficient of Determination is the second regression-based error metric, and is the proportion of variance explained by the model, The R^2 is defined in Equation 20:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \quad (20)$$

where y_i is a response variable, $\hat{y}_i$ is the corresponding estimated response, and $\bar{y}$ is the mean of the response variables.

Finally, the Mean Square Error (MSE) is a direct measure of the error produced by the model, with the lower the value the better, and is defined in Equation 21:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2, \quad (21)$$

where N is the number of observations, y_i is the measured response, and $\hat{y}_i$ is the estimated response.

3.3.2. Classification Error Metrics

Traditionally in ML classification settings, the Accuracy (defined in Equation 22), Precision, Recall and F1 score are reported. We also investigated the Normalised Mutual Information, which suggests how dependent one set of data is upon another. We do not report the Precision, Recall, F1 Score and Normalised Mutual Information, however, as we found they provided no additional information.

It is important to note that the main error metric being compared and minimised in both the regression and classification tests is the $\eta_{0.15}$ outlier rate. As this outlier rate is explicitly accepting of a level of inaccuracy that scales with redshift, there is an inherent acceptance of some level of leakage between neighbouring classes during classification tests.

This acceptance of leakage means that models may present with high error rates due to rigid correct/incorrect classifications, and yet may still be acceptable models for this process.

$$\text{Accuracy}(y, \hat{y}) = \frac{1}{n} \sum_{i=0}^{n-1} I(\hat{y}_i = y_i), \quad (22)$$

where y is the measured response, $\hat{y}$ is the predicted response, n is the number of samples, and I is an indicator function, indicating the cases where the predicted response matched the measured response.

3.4. Statistical Significance

Analysis of Variance (ANOVA) tests were used in this study to test for statistical significance. This allows us to test whether the changes in model correlate to a statistically significant change in estimated redshift, by testing to see if the means of two or more populations (in this case, experiments with different models) differ.

In all cases, the tests were run in a one-vs-many scenario, with the one being tested as our best performing metric in order to determine whether our best performing result is statistically significant.

3.5. Software

This work makes use of the Scikit-learn³ Python package (Pedregosa et al., 2011) for the implementation of the RF and kNN algorithms, as well as the Euclidean and Mahalanobis distance metrics. We made use of the PyLMNN⁴ package for the LMNN distance metric, and the metric-learn⁵ package for the MLKR distance metric. The code and data used in this work is available on Github⁶.

4. Results

Given the number of different algorithms, distance metrics and datasets tested in this work, we have assigned each combination a Test ID, defined in Table 2 for Regression-based tests, and Table 3 for Classification-based tests. Each test ID is made up of 4 characters, with the first character (R/C) representing whether the test is a regression- or classification-based test, the 2nd and 3rd characters (Eu/Ma/ML/Rf) representing the method used for estimation (Euclidean distance, Mahalanobis distance, a Learned distance metric, or Random Forest), and the final character (1/2/3) representing the training set used (1 = Random, 2 = ELAIS-S1, 3 = eCDFS).

As discussed in Section 2, we tested both statistical standardisation of input photometry, compared with the use of astrophysically derived “colours”, and taking the log of radio and infrared data to better distribute the data. In both cases, however, the result was very similar to the standardised dataset, and hence is the only value quoted.

³<https://scikit-learn.org/>

⁴<https://pypi.org/project/PyLMNN/>

⁵http://contrib.scikit-learn.org/metric-learn/generated/metric_learn.MLKR.html

⁶<https://github.com/kluken/Redshift-kNN-2021>

Table (2) Description of the regression based tests conducted. The Test ID is used to reference the results in Table 4. The Method is the ML algorithm used in the test. The Distance Metric is the distance metric used in the kNN algorithm (and hence is empty for the RF algorithm). The Training Set is the method of choosing the training set for the ML algorithm.

Test ID	Method	Distance Metric	Training Set
REu1	kNN	Euclidean	Random
REu2	kNN	Euclidean	ELAIS-S1
REu3	kNN	Euclidean	eCDFS
RMa1	kNN	Mahalanobis	Random
RMa2	kNN	Mahalanobis	ELAIS-S1
RMa3	kNN	Mahalanobis	eCDFS
RML1	kNN	MLKR	Random
RML2	kNN	MLKR	ELAIS-S1
RML3	kNN	MLKR	eCDFS
RRf1	RF	–	Random
RRf2	RF	–	ELAIS-S1
RRf3	RF	–	eCDFS

Table (3) Description of the classification based tests conducted. The Test ID is used to reference the results in Table 5. The Method is the ML algorithm used in the test. The Distance Metric is the distance metric used in the kNN algorithm (and hence is empty for the RF algorithm). The Training Set is the method of choosing the training set for the ML algorithm.

Test ID	Method	Distance Metric	Training Set
CEu1	kNN	Euclidean	Random
CEu2	kNN	Euclidean	ELAIS-S1
CEu3	kNN	Euclidean	eCDFS
CMa1	kNN	Mahalanobis	Random
CMa2	kNN	Mahalanobis	ELAIS-S1
CMa3	kNN	Mahalanobis	eCDFS
CML1	kNN	LMNN	Random
CML2	kNN	LMNN	ELAIS-S1
CML3	kNN	LMNN	eCDFS
CRf1	RF	–	Random
CRf2	RF	–	ELAIS-S1
CRf3	RF	–	eCDFS

Table (4) Results using the regression based algorithms. The Test column relates to the tests defined in Table 2. The best result for each metric is highlighted in bold.

Test	k / Trees	$\eta_{0.15}$ (%)	$\eta_{2\sigma}$ (%)	R^2 Value	MSE	σ	σ_{NMAD}
REu1	5	8.14	6.61	0.65	0.06	0.10	0.05
REu2	5	8.70	5.51	0.60	0.06	0.12	0.06
REu3	11	8.28	4.85	0.53	0.09	0.12	0.06
RMa1	5	5.85	5.09	0.59	0.07	0.12	0.03
RMa2	3	8.70	5.39	0.58	0.07	0.15	0.04
RMa3	5	7.68	5.45	0.54	0.09	0.13	0.04
RML1	4	6.36	4.07	0.73	0.05	0.10	0.05
RML2	9	8.58	4.78	0.68	0.05	0.11	0.06
RML3	5	10.91	4.44	0.58	0.08	0.13	0.05
RRf1	44	8.14	4.84	0.59	0.07	0.13	0.05
RRf2	21	12.75	5.88	0.32	0.11	0.19	0.06
RRf3	23	10.71	4.65	0.54	0.09	0.14	0.06

4.1. Regression

Based on the Regression tests outlined in Table 2, we present the results in a series of plots (Figures 9, 10, 11, and 12).

In each plot three subfigures are presented, with Subfigure A representing the random training set, Subfigure B representing the training set built using galaxies from the ELAIS-S1 field, and Subfigure C representing the training set built using galaxies from the eCDFS field. The subfigures are split into two plots – the top plot showing the measured spectroscopic redshift (x-axis) compared with the predicted redshift (y-axis), with a perfect 1:1 correlation (red dashed line) and the $\eta_{0.15}$ outlier rate boundaries (blue dashed lines) shown. The bottom plot shows the measured spectroscopic redshift (x-axis) against the residuals, again with the perfect 1:1 correlation (red dashed line) and the $\eta_{0.15}$ outlier rate boundaries (blue dashed lines) shown.

These results are summarised in Table 4.

We show that the lowest $\eta_{0.15}$ outlier rate is achieved using the kNN algorithm paired with the Mahalanobis distance metric, and is statistically different from most other algorithms (kNN using Euclidean Distance: p value = 0.0183, and the RF algorithm: p value = 0.0183 compared with kNN using the Mahalanobis distance metric as a baseline). The kNN algorithm using the MLKR distance metric (a Mahalanobis-like distance metric) is not statistically significantly different (p value = 0.5750). However, all results (including the RF algorithm) suffer from the same issues of under-predicting high redshift ($z > 1$) galaxies.

The randomly selected training sets typically achieve lower $\eta_{0.15}$ outlier rates than the training sets built using one field only, however, neither field-based training set provides consistently better $\eta_{0.15}$ outlier rates than the other. This is confirmed statistically, with no statistically significant result measured (p value = 0.2072).

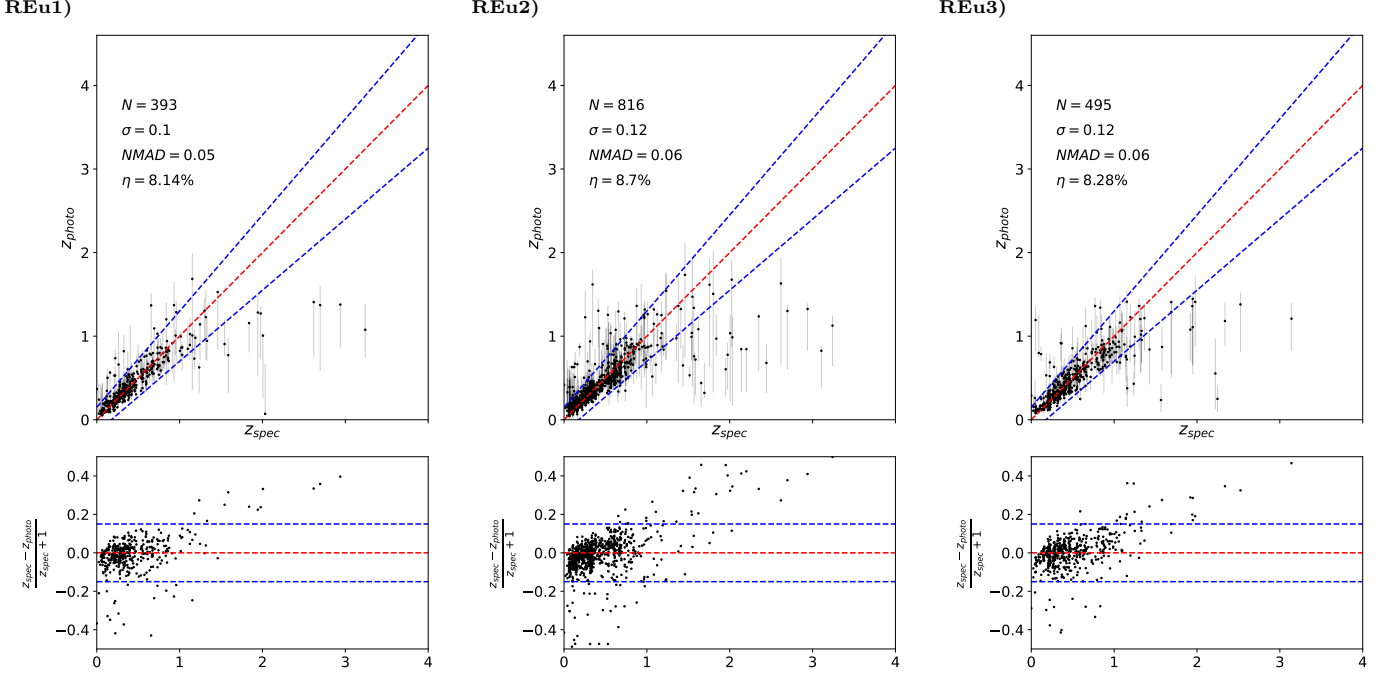


Figure (9) Figures showing the results using the kNN Regression algorithm using Euclidean Distance as the Distance metric, varying the data used for training (with the numbers corresponding to the Test ID in Table 2). (REu1 – Left) uses a random training sample, (REu2 – centre) uses the ELAIS-S1 field as the training set, and (REu3 – right) uses the eCDFS field as the training set. The x-axes shows the measured spectroscopic redshift. The top-panel y-axes show the predicted redshift using the given model. The bottom-panel y-axes show the normalised residuals. The red-dashed line shows a perfect 1:1 prediction, and the blue-dashed lines show the decision boundaries based on the $\eta_{0.15}$ outlier rates.

4.2. Classification

Based on the Classification tests outlined in Table 3, we show the classification-based results for the kNN algorithm – using Euclidean distance (Figure 14), Mahalanobis distance (Figure 15) and the LMNN learned distance metric (Figure 16) – and the RF algorithm (Figure 17), summarised in Table 5.

We present our classification-based results using **scaled confusion matrices**, where the x-axis shows the measured spectroscopic redshift, the y-axis shows the predicted redshift, the colour shows the density of the objects in that chosen bin, and the width of each bin is proportional to the range of redshift values represented by the bin. An example scatter plot (in the same style as Figure 9), demonstrating the effect of binning on the classification results when compared against the original spectroscopic measurements is shown in Figure 13.

The kNN algorithm paired with the Mahalanobis distance metric provides the lowest $\eta_{0.15}$ and $\eta_{2\sigma}$ outlier rates, as well as performing the best in terms of traditional ML classification metrics (accuracy, precision, recall and F1 score). However, while the results using the Mahalanobis distance metric are statistically significantly better than those using the Euclidean distance metric ($p=0.00619$), they are not statistically different from the LMNN learned distance metric (a Mahalanobis-like distance metric; $p=0.4276$),

Table (5) Results using the classification based algorithms. The Test column relates to the tests defined in Table 3. The best result for each metric is highlighted in bold

Test	k / Trees	$\eta_{0.15}$ (%)	$\eta_{2\sigma}$ (%)	Acc	σ
CEu1	6	8.91	4.07	0.30	0.10
CEu2	5	12.13	3.55	0.24	0.14
CEu3	8	11.31	5.05	0.27	0.13
CMa1	11	5.85	3.57	0.50	0.14
CMa2	6	7.60	4.41	0.37	0.14
CMa3	13	7.88	3.43	0.40	0.13
CML1	7	6.36	5.85	0.40	0.09
CML2	8	9.93	4.04	0.32	0.16
CML3	7	9.90	4.04	0.40	0.13
CRf1	53	7.12	3.82	0.36	0.10
CRf2	39	10.54	4.41	0.27	0.14
CRf3	32	10.91	4.24	0.33	0.15

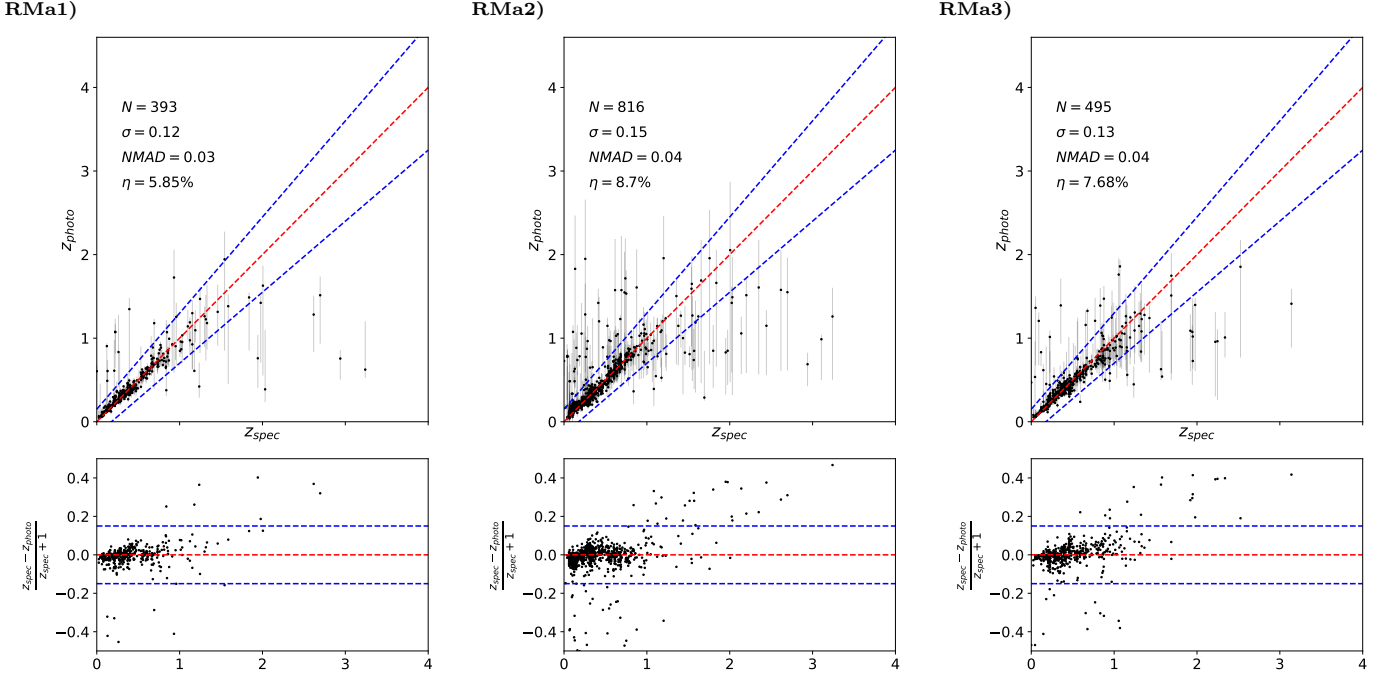


Figure (10) Figures showing the results using the kNN Regression algorithm using Mahalanobis Distance as the Distance metric, varying the data used for training (with the numbers corresponding to the Test ID in Table 2). (RMa1 – Left) uses a random training sample, (RMa2 – centre) uses the ELAIS-S1 field as the training set, and (RMa3 – right) uses the eCDFs field as the training set. The x-axes shows the measured spectroscopic redshift. The top-panel y-axes show the predicted redshift using the given model. The bottom-panel y-axes show the normalised residuals. The red-dashed line shows a perfect 1:1 prediction, and the blue-dashed lines show the decision boundaries based on the $\eta_{0.15}$ outlier rates

or the RF algorithm ($p=0.8913$). Again, we note that for most algorithms, the highest redshift galaxies remain a problem for estimation, however, the results can change significantly based on the distance metric used – for example, the $z > 1.02$ bin was correctly predicted $\sim 50\%$ of the time using the kNN algorithm paired with Euclidean distance, however, using the Mahalanobis distance metric brought this up to $\sim 74\%$.

As with the regression tests, the random training sample outperformed the training sets built from a single field in the primary $\eta_{0.15}$ error metric, however, the results are statistically insignificant ($p=0.4397$).

5. Discussion

Most previous work has treated the estimation of galaxy redshift as a regression task – estimating a continuous value for redshift. We have shown that the kNN algorithm – when using either a learned distance metric, or the Mahalanobis distance metric – outperforms the RF algorithm, particularly at a redshift of $z < 1$. At a redshift of $z > 1$, both the kNN and RF have systematic under-estimations, likely caused by the unbalanced training sample used, with the vast majority of samples at lower redshifts.

Here we try a different approach, by treating the problem of calculating point estimates of redshifts as a classifi-

cation problem. Previous works have calculated probability density functions (PDFs) for galaxies by finely binning data and treating it as a classification problem (Gerdes et al., 2010; Pasquet-Itam and Pasquet, 2018; Eriksen et al., 2020). However, this is subtly different to this work as the primary purpose here is to identify the high-redshift galaxies – not to generate PDFs. We balance the data by binning the data into 15 bins with equal numbers of sources. By allowing this coarse mapping, we ensure that all redshift values we are attempting to predict are equally represented, and can therefore obtain better results at the higher redshift ranges. In particular, we have included a $z > 1$ bin, which in our best results (the kNN algorithm using the Mahalanobis distance metric) we predict in 74% of cases (85% of galaxies at $z > 0.8$ predicted in the highest two bins). Again, the kNN algorithm is able to outperform the RF algorithm.

In addition to the regression and classification tests, we tested using different training samples. The first training sample is made up of a random selection across both the ELAIS-S1 and eCDFs fields. The second and third samples were made up of only one of the fields, which was then used to predict the other. As expected, the random sample modelled the test sets better – though not significantly – than the training samples taken from a single field only. This is likely due to observational differences – while the

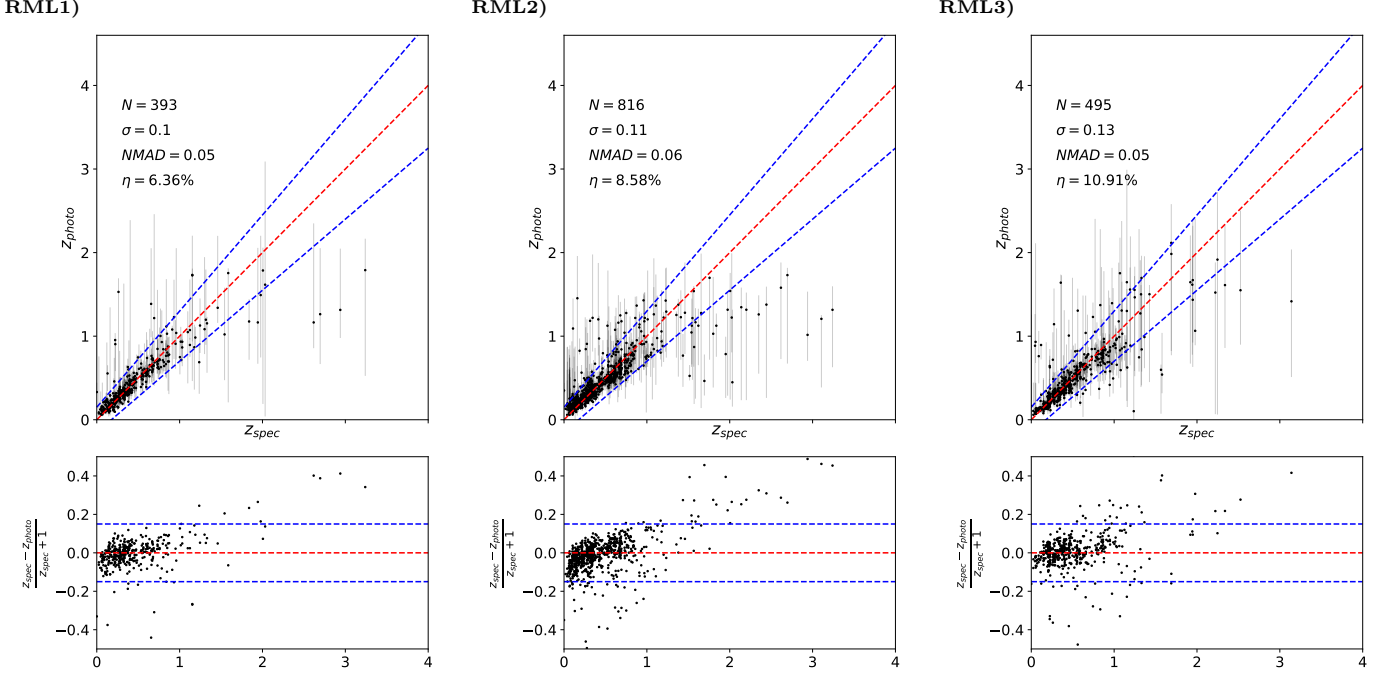


Figure (11) Figures showing the results using the kNN Regression algorithm using MLKR Distance as the Distance metric, varying the data used for training (with the numbers corresponding to the Test ID in Table 2). (RML1 – Left) uses a random training sample, (RML2 – centre) uses the ELAIS-S1 field as the training set, and (RML3 – right) uses the eCDFS field as the training set. The x-axes shows the measured spectroscopic redshift. The top-panel y-axes show the predicted redshift using the given model. The bottom-panel y-axes show the normalised residuals. The red-dashed line shows a perfect 1:1 prediction, and the blue-dashed lines show the decision boundaries based on the $\eta_{0.15}$ outlier rates.

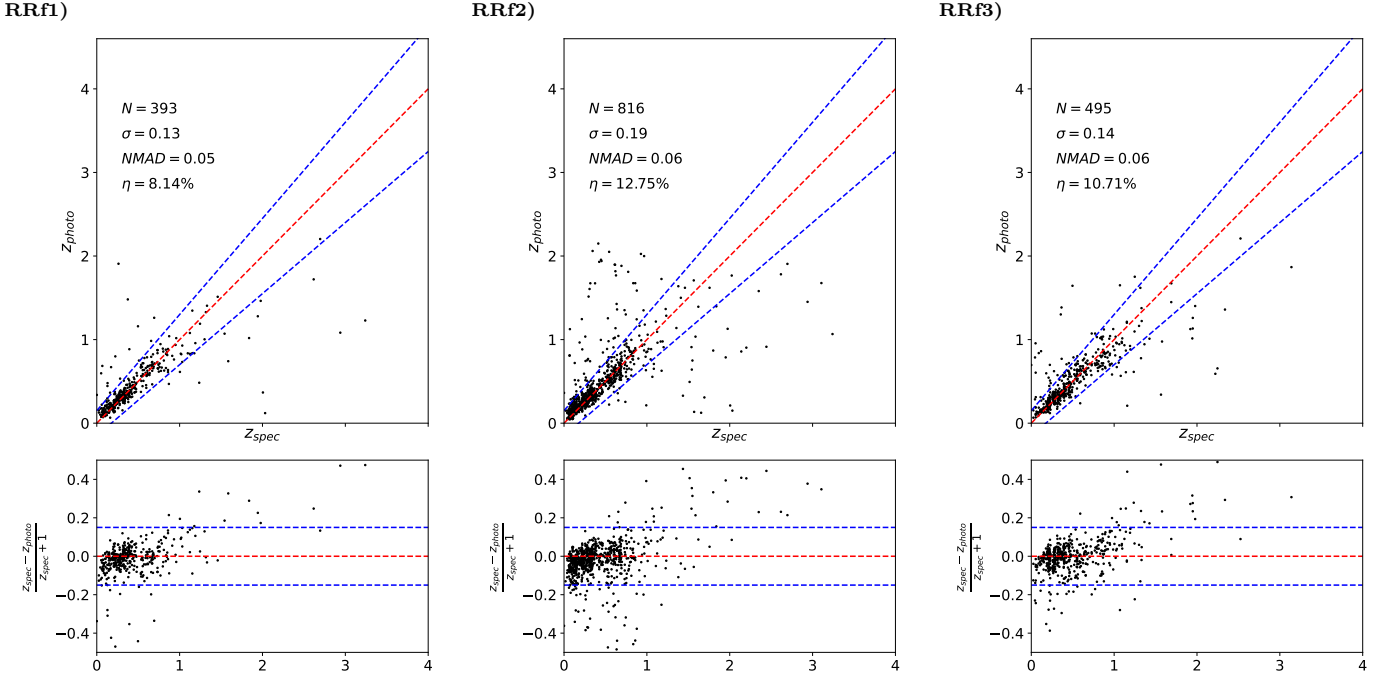


Figure (12) Figures showing the results using the RF Regression algorithm, varying the data used for training (with the numbers corresponding to the Test ID in Table 2). (RRf1 – Left) uses a random training sample, (RRf2 – centre) uses the ELAIS-S1 field as the training set, and (RRf3 – right) uses the eCDFS field as the training set. The x-axes shows the measured spectroscopic redshift. The top-panel y-axes show the predicted redshift using the given model. The bottom-panel y-axes show the normalised residuals. The red-dashed line shows a perfect 1:1 prediction, and the blue-dashed lines show the decision boundaries based on the $\eta_{0.15}$ outlier rates.

surveys used in this work were designed to be as homogeneous as possible, there will always be slight differences.

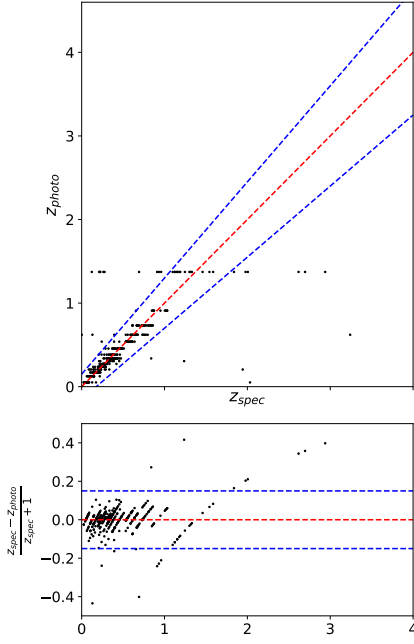


Figure (13) An example scatter plot similar to Figure 9 for the CMA1 test, showing the effect of binning and classifying.

One such difference is the number of sources in each field. While there is not expected to be differences in the actual source counts between these fields, there is the potential issue of minor observational issues (for example, in the radio regime, there may have been more radio-frequency interference in one set of observations than another. In the optical regimes, it may have been that there was slightly different sky conditions affecting the observations) affecting the fields differently.

5.1. Comparison with Previous Works

Fair comparisons with previous works are often difficult due to different features being selected — often studies are conducted on well-observed, feature rich fields like the COSMOS field, providing a wealth of UV, NIR and X-ray data that is typically not available for the majority of the sky — and different selection methods — selecting sources from the SDSS Galaxy survey, where the redshift range is restricted towards higher redshift and typically don't have a radio-counterpart, which are shown by Norris et al. (2019) to be a more difficult challenge. Where the data sets have been AGN selected, using similar features as in Duncan et al. (2021, Lockman Hole AGN Experiments), we find our results are highly competitive, providing a lower outlier rate ($\sim 6\%$ compared with $\sim 22\%$, albeit with a higher scatter ($\sigma = 0.12$ compared with $\sigma = 0.077$), suggesting that while we are less accurate than others, we are having fewer estimates catastrophically fail. This optimisation of outlier rate (reducing the number of estimations that catastrophically fail) is motivated by the science goals of the EMU Project. We note that this comparison should be further tempered by the fact that the differing training

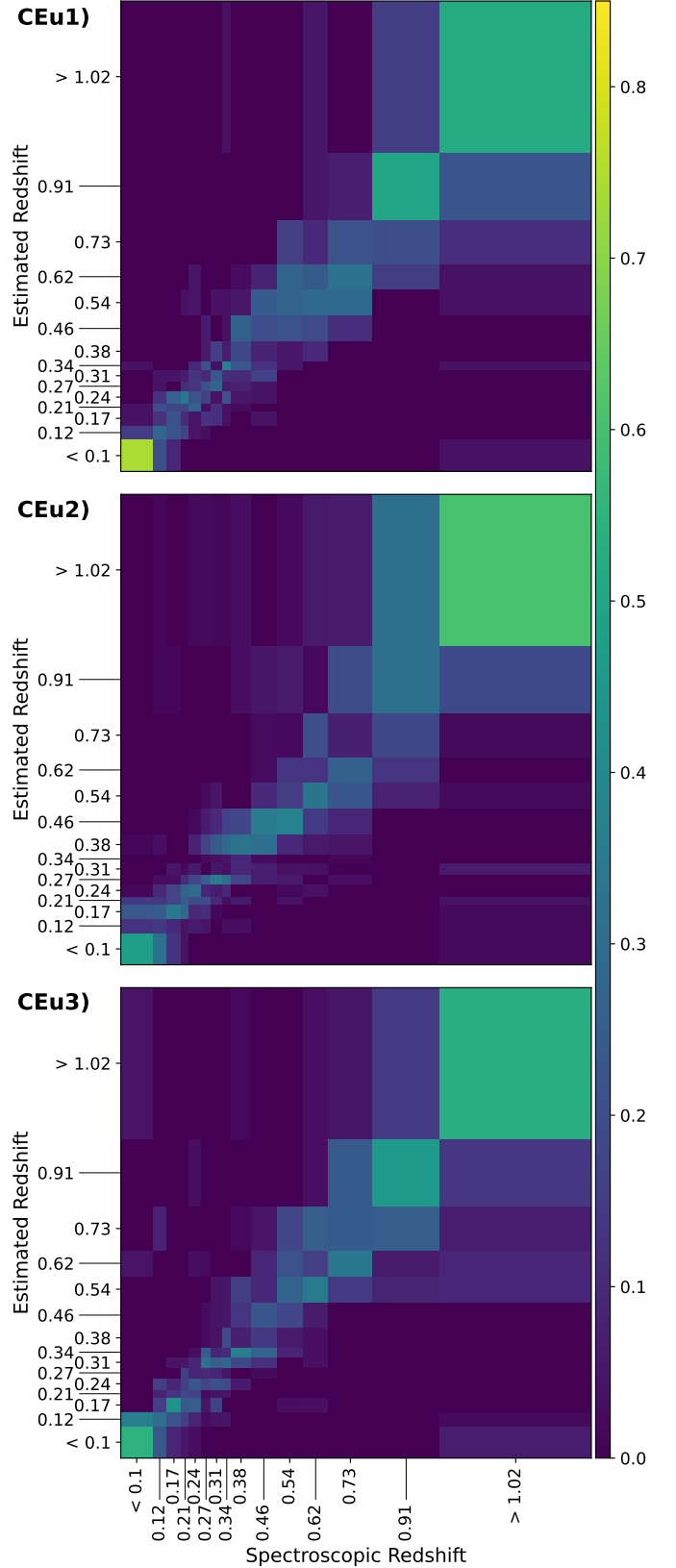


Figure (14) The confusion matrix of the kNN classification tests using the Euclidean distance metric. The x-axis of each plot shows the measured spectroscopic redshift, and the y-axis shows the estimated redshift. The size of the boxes is scaled based on the width of the bin used for classification. (CEu1 – Top) uses a random training sample, (CEu2 – Middle) uses a training sample consisting of the ELAIS-S1 field, and (CEu3 – Bottom) uses a training sample consisting of the eCDFs field.

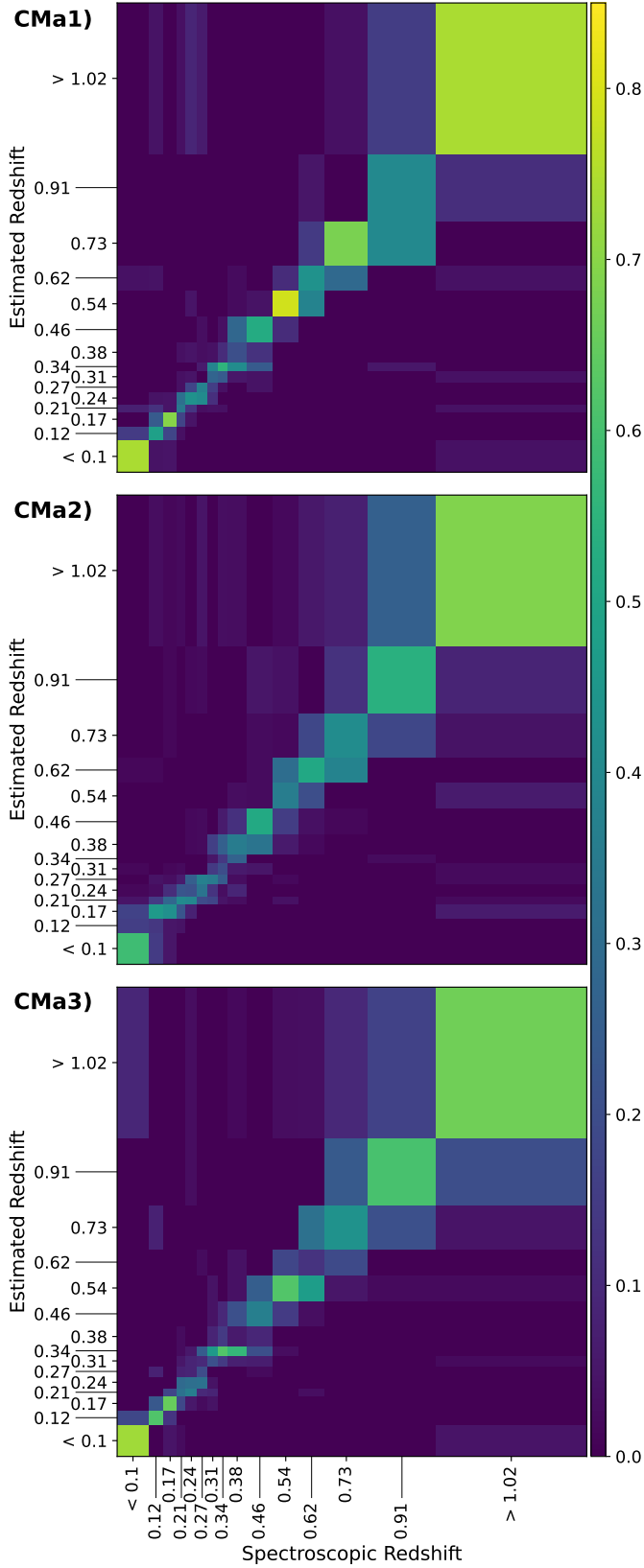


Figure (15) The confusion matrix of the kNN classification tests using the Mahalanobis distance metric. The x-axis of each plot shows the measured spectroscopic redshift, and the y-axis shows the estimated redshift. The size of the boxes is scaled based on the width of the bin used for classification. (CMa1 – Top) uses a random training sample, (CMa2 – Middle) uses a training sample consisting of the ELAIS-S1 field, and (CMa3 – Bottom) uses a training sample consisting of the eCDFs field.

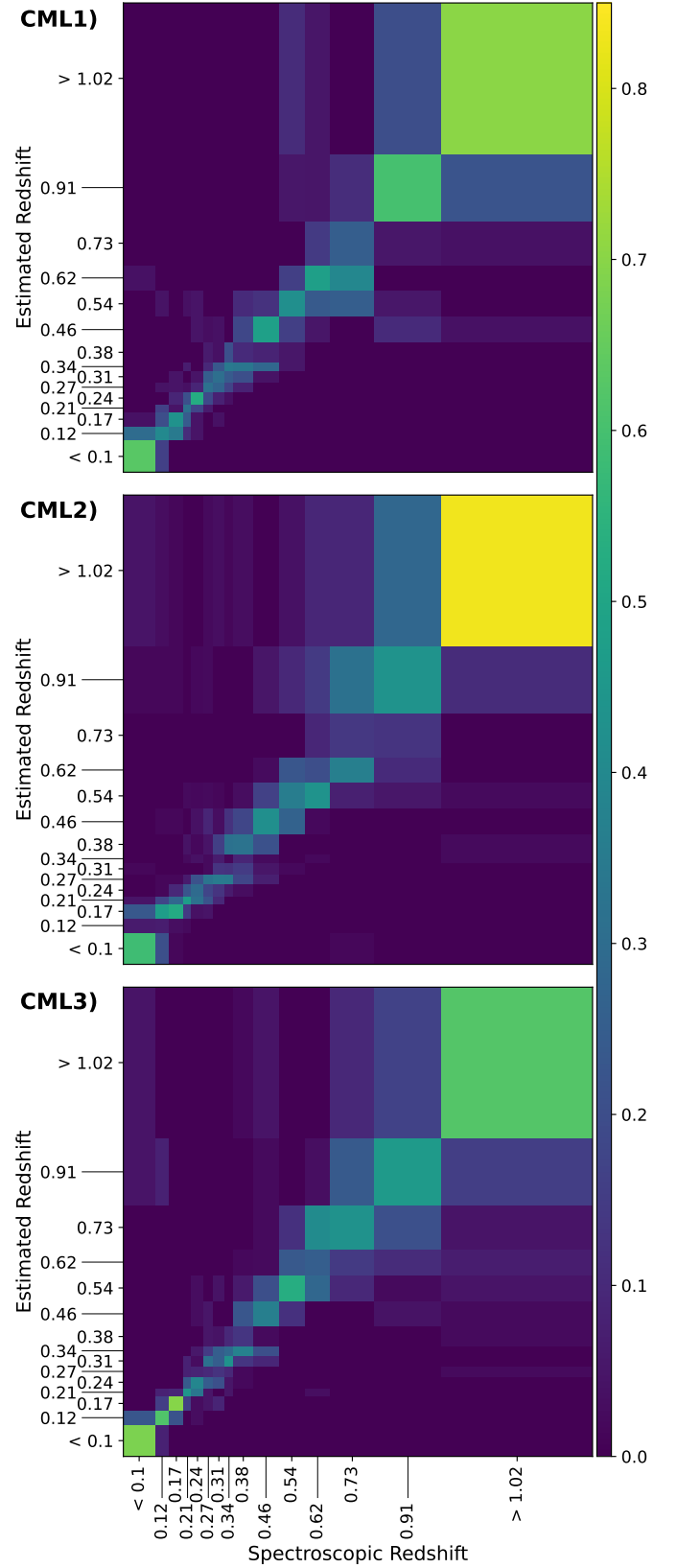


Figure (16) The confusion matrix of the kNN classification tests using the LMNN learned distance metric. The x-axis of each plot shows the measured spectroscopic redshift, and the y-axis shows the estimated redshift. The size of the boxes is scaled based on the width of the bin used for classification. (CML1 – Top) uses a random training sample, (CML2 – Middle) uses a training sample consisting of the ELAIS-S1 field, and (CML3 – Bottom) uses a training sample consisting of the eCDFs field.

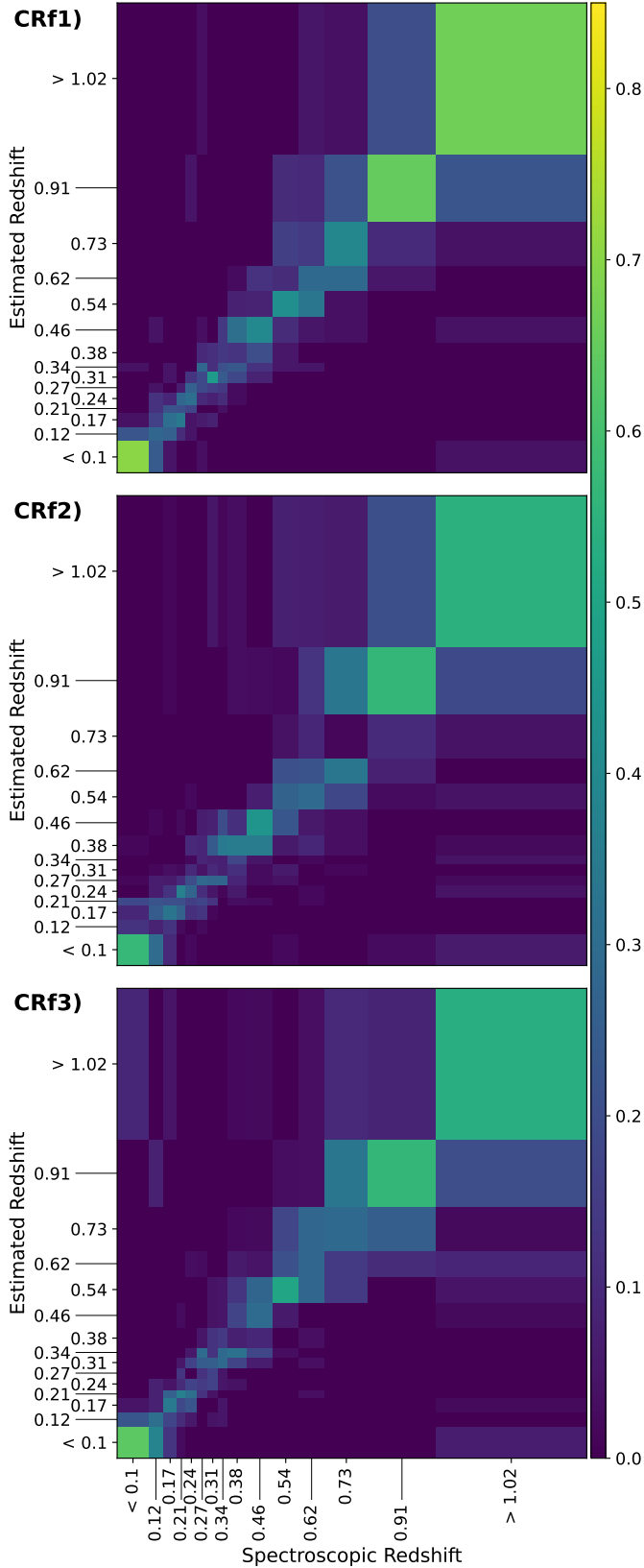


Figure (17) The confusion matrix of the RF classification tests. The x-axis of each plot shows the measured spectroscopic redshift, and the y-axis shows the estimated redshift. The size of the boxes is scaled based on the width of the bin used for classification. (CRf1 – Top) uses a random training sample, (CRf2 – Middle) uses a training sample consisting of the ELAIS-S1 field, and (CRf3 – Bottom) uses a training sample consisting of the eCDFS field.



Figure (18) A comparison of the $\eta_{0.15}$ outlier rates presented in Section 4. The top panel compares the regression-based tests, with the bottom panel comparing the classification-based tests. The x-axis shows the different tests run, with the y-axis showing the $\eta_{0.15}$ outlier rate. The different colours/styled lines represent the different training data used – the purple solid line used a training sample randomly selected from both the ELAIS-S1 and eCDFS fields, the blue dotted line used a training sample comprised of galaxies in the ELAIS-S1 field, and the green dashed line used a training sample comprised of galaxies in the eCDFS field.

and test set sizes, and the redshift distribution of the overall datasets have a large impact on the overall statistics.

6. Conclusion

We have used a radio-selected point-source catalogue consisting of ATLAS radio data, DES optical photometry, SWIRE infrared data and OzDES spectroscopic redshifts to test the kNN and RF algorithms for estimating the redshift of galaxies.

We have used both regression and classification modes of these algorithms in order to balance highly unbalanced training sets, and have shown that using classification modes increases the effectiveness of the algorithms at a redshift of $z > 1$ (noting that while we are generally able to identify the galaxies at $z > 1$, we are unable to estimate their exact redshift). Given this increase in effectiveness at isolating the high-redshift galaxies using classification methods, a mix of classification- and regression-based methods would provide the best possible result over larger redshift ranges, allowing the higher redshift galaxies to be identified, while still being able to estimate the redshift of nearby galaxies to high accuracy.

In our tests we have shown that when using the classification modes of the algorithms, the kNN algorithm using the Mahalanobis distance metric performs better than the alternative methods tested. We particularly note that the $z > 1.02$ bin was correctly predicted in 74% of cases – far better than the regression regimes that fail consistently at that range.

When using the regression modes of the algorithms, the kNN algorithm using the Mahalanobis distance metric performed statistically significantly better than most of the alternate methods tested (kNN using Euclidean Distance: p value = 0.0183, and the RF algorithm 0.0183) – the kNN algorithm paired with the Mahalanobis-like MLKR distance metric was statistically insignificantly different (p = 0.5750). In both regression and classification methods, the kNN algorithm outperforms the much more widely used RF algorithm.

Finally, we tested whether there would be a significant difference in the $\eta_{0.15}$ outlier rate between a model trained and tested on randomly split data, and a model that is trained on one field of sky and tested on another. While the results were generally suggestive of the random training sample out-performing the field-specific training samples, the difference is not statistically significant (Regression p value = 0.2072 and classification p value = 0.4397). This suggests that for new fields being observed with similar strategies to previously observed fields, any differences in measured photometry should produce minimal effect on the estimated redshifts. It should be noted, however, that this does not take into consideration any differences in photometry caused by differences in telescopes. For example, if a galaxy was to have SkyMapper g , r , i and z photometry instead of DES photometry. To determine the impact of changes of telescopes on the data would require further testing, and will be the subject of further work utilising Transfer Learning.

7. Implications for Future Radio Surveys

Traditional photometric template fitting methods typically struggle to estimate the redshift of radio galaxies. In the near future, large area radio surveys like the EMU survey are set to revolutionise the field of Radio Astronomy, with the number of known radio galaxies set to increase by an order of magnitude. This work shows that broadband photometry at similar wavelengths to those available in present and near-complete all-sky surveys will be enough to estimate acceptable redshifts for $\sim 95\%$ of radio sources with coverage over those bands. Further, while continuous redshift values are particularly difficult to estimate for radio sources at high-redshift, they can still be identified as high redshift with the majority of sources correctly placed in the highest redshift bin using classification modes.

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